

Other representations (product display, app, etc) of the onboarding material SHOULD follow the [Matter Brand Guidelines](#).

5.1.3. QR Code

The Onboarding Payload MAY be included on (or with) a device in the form of a QR code. The following sections detail the content, encoding, and formatting of the QR code.

5.1.3.1. Payload

The content of the QR code consists of the concatenation of a prefix string and a Base-38-encoded string containing the required and optional TLV content:

QR code string := <prefix><base-38-content>

Prefix String

The prefix string consists of the three-character string:

MT:

Base-38 Content

The required content of the QR code is composed of a Packed Binary Data Structure containing elements of the Onboarding Payload as detailed below. The resulting data is Base-38 encoded (with a specific alphabet) to form a string compatible with alphanumeric QR encoding.

Packed Binary Data Structure

Individual data elements SHALL be placed into the structure in the order detailed in the table below. Elements being packed are not necessarily byte- or word-aligned. The resulting packed structure is presented to the encoder as a multi-byte array (see [Section 5.1.3.1.5, “Encoding”](#)), which SHALL be padded with '0' bits at the end of the structure to the nearest byte boundary.

The bits of each fixed-size value are placed in the packed binary data structure in order from least significant to most significant. If TLV Data is included, it is appended to the end of the packed binary data.

For example, the first elements in the table below SHALL be packed into the first bytes of the data array as pictured:

Table 56. Packing of the onboarding payload

lsb		Byte 0					msb		Byte 1					Byte 2					Byte 3					...				
0							7	0								7	0							7	0			
version			Vendor ID												Product ID													
0		2	0												15	0												15